

QUESTIONS FOR CHAPTER 3—PROJECT INITIATION

问题：第三章—项目启动

1. There are many **contractual arrangements** for handling a project. This chapter describes the arrangements in four basic forms: **design-bid-build**, **design-build**, **construction management**, and **owner-agent**. Describe the **advantages** and **disadvantages** of each of these arrangements, considering factors such as **cost**, **time**, and the **level of involvement** desired by the **owner** of the project.

关于项目管理，有多种**合同安排**。本章描述了四种基本形式的安排：**设计-投标-建造**、**设计-建造**、**施工管理**和**业主代理**。描述每种安排的**优点**和**缺点**，并考虑**成本**、**时间**以及**业主期望的参与程度**等因素。

(1) Design/Bid/Build

Advantages:

- The **cost** of the project is known before commitment to signing **construction contracts**.
- The **responsibility**, **risk**, and **involvement** of all parties are well-defined.
- The **owner** has a high level of **involvement** and **control** during **design**.
- Selected for projects when **cost** is primary, **schedule** is secondary, and **scope** is well-defined.

Disadvantages:

- **Changes** made after awarding **construction contracts** can be costly.
- Major changes outside the original **contract bid documents** may be difficult and costly.
- The **time** required for the **design**, **bid**, and **construction** process can be extensive.
- The **owner** has little **control** over the project during **construction**.

设计-投标-建造 (Design/Bid/Build)

优点:

- 在签订**施工合同**前，项目的**成本**是已知的。
- 各方的**责任**、**风险**和**参与程度**明确。
- **业主**在**设计**阶段拥有较高的**参与**和**控制**。
- 适用于**成本**优先、**进度**次要、**范围**明确的项目。

缺点:

- 在**施工合同**签订后进行的**变更**可能会很昂贵。
- 原始**合同投标文件**之外的重大变更可能会困难且昂贵。
- **设计**、**投标**和**施工**过程所需的**时间**可能会很长。
- **业主**在**施工**阶段对项目几乎没有**控制权**。

简化版:

Design/Bid/Build	设计-投标-建造 (Design/Bid/Build)
Advantages: • Cost known upfront • Responsibilities clearly defined • Owner control in design	优点: • 成本提前确定 • 责任明确 • 业主控制设计过程
Disadvantages: • Changes are costly • Time-intensive process Owner control limited during construction	缺点: • 变更成本高 • 时间长 • 施工时业主控制少

(2) Design/Build

Advantages:

- **Time of completion** is usually reduced because **construction** can start before 100% of the **design** is completed.
- Reduces **adversarial relationships** between **design** and **construction personnel**.
- Provides **flexibility** to the **owner** for revising the **design** during **construction**.
- Having one point of **contact** can reduce the **owner's workload** if the **project scope** is clear.

Disadvantages:

- **Costs** to the **owner** may be increased because **competitive bidding** is precluded.
- The total **project cost** is not well-known at the beginning of the project.
- **Inspection management** can be challenging since the **designer** is also the **builder**.
- Considerable **time** is required of the **owner** to assist in **major decisions** and monitor **costs**.

设计-建造 (Design/Build)

优点:

- 完工时间通常较短，因为在设计完成 100%之前即可开始施工。
- 减少了设计与施工人员之间的对立关系。
- 为业主在施工过程中调整设计提供了灵活性。
- 如果项目范围明确，一个联系人可以减少业主的工作量。

缺点:

- 由于排除了竞争性投标，业主的成本可能会增加。
- 项目的总成本在项目开始时并不清楚。
- 检查管理可能会有挑战，因为设计师同时是建造方。
- 业主需要花费大量时间协助重大决策和监控成本。

简化版:

Design/Build Advantages: • Shorter time to complete • Reduces conflicts (between design and construction teams) • Flexibility to adjust design Disadvantages: • Higher cost possible (no competitive bidding) • Total cost unknown at start • Inspection is challenging (designer is builder)	设计-建造 (Design/Build) 优点: • 时间更短 • 减少冲突（设计和施工人员） • 灵活调整设计 缺点: • 成本可能更高 • 总成本不清楚（项目开始时） • 检查需特别关注（设计师也是施工方）
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(3) Construction Management

Advantages:

- **Single-source management** allows the **owner** to control their level of **involvement**.
- Potential **cost reductions** by eliminating **contract administration** costs from **contractors**.
- Potential **time savings** by integrating all parties from the start to the end of the project.
- Reduces **adversarial relationships** by fostering **teamwork** among parties.
- Allows the **owner** to focus on the **overall project aspects** rather than **daily issues**.

Disadvantages:

- **Multiple construction contracts** may lead to **gaps** or **overlaps** in work if not monitored.
- Tracking the **costs** of multiple **construction packages** may be challenging.
- **Delays** can occur due to **contractors** interfering with each other's work on-site.
- The **owner's involvement** may diminish as the **number of contracting parties** increases.
- The **construction manager (CM)** must be brought on board early to utilize their **construction knowledge**.

施工管理 (Construction Management)

优点:

- 单一来源管理允许业主控制其参与程度。
- 通过减少来自承包商的合同管理费用，可能实现成本降低。
- 通过从项目开始到结束的整合所有方，可能实现时间节省。
- 通过促进各方之间的团队合作，减少对立关系。
- 使业主可以专注于项目整体方面，而不是日常问题。

缺点:

- 多个施工合同可能会导致工作重叠或间隙，如果不加以监控。
- 跟踪多个施工包的成本可能会很困难。
- 由于承包商在施工现场的工作干扰，可能会出现延误。
- 随着合同方数量的增加，业主的参与度可能会减少。
- 必须尽早引入施工经理 (CM)，以便利用他们的施工知识。

简化版:

Construction Management Advantages: • Single management source, strong owner control • Cost savings (less contract admin costs) • Time savings (teamwork among parties) Disadvantages: • Multiple contracts, risk of overlap/gaps • Cost tracking is difficult • Interference risk on-site	施工管理 (Construction Management) 优点: • 单一管理，业主控制强 • 节省成本（减少合同管理费用） • 节省时间（团队合作） 缺点: • 合同多，需防重叠/间隙 • 成本追踪难 • 干扰风险高（施工现场）
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(4) Owner/Agent

Advantages:

- Better **cost control** since the project is managed **in-house** by the **owner**.
- Simplified **schedule maintenance** as the **owner** acts as the **manager**.
- The **owner** remains involved throughout all **project phases**.
- **Priorities** and **quality standards** are well-defined.
- This arrangement is generally used by **owners** with substantial **construction knowledge**.

Disadvantages:

- **Cost control** may fall under **purchasing** instead of **engineering**.
- **Contract scheduling** may be affected by other projects within the **owner's organization**.
- Involvement of **in-house engineering** may be lost during **construction**.
- **Gaps** or **overlaps** in work can occur unless a single person oversees all **phases** from **design** to **construction**.

业主/代理 (Owner/Agent)

优点:

- **成本控制**更好，因为项目由**业主**内部管理。
- **进度管理**简化，因为**业主**是**管理者**。
- **业主**在整个项目阶段中都有参与。
- 关于**优先级**和**质量标准**的问题得到明确定义。
- 此安排通常适用于具备丰富**施工知识**的**业主**。

缺点:

- **成本控制**可能归于**采购**部门，而非**工程**部门。
- 其他项目可能会影响**业主组织**内的**合同进度**。
- **内部工程**部门在**施工**过程中可能失去参与。
- 如果没有一个人从**设计**到**施工**全面跟进各个**阶段**，可能会出现**工作重叠**或**间隙**。

简化版:

Owner/Agent	业主/代理 (Owner/Agent)
Advantages: • Stronger cost control (managed in-house) • Better schedule control (owner as manager) • Owner involvement throughout project	优点: • 成本控制更强（业主内部管理） • 进度控制更好（业主亲自管理） • 业主参与项目全过程
Disadvantages: • Cost control may focus on purchasing • Impact from owner's other projects • Less engineering team involvement	缺点: • 成本控制偏向采购 • 受业主其他项目影响 • 工程团队参与少

2. Review one of the **references** at the end of this chapter, or choose another reference, and briefly describe the many forms of **construction management** that are currently being used in the **engineering and construction industry**.

查看本章末尾的**参考文献**之一，或选择其他参考文献，简要描述目前在**工程**和**建筑行业**中使用的多种形式的**施工管理**。

(1) Agency Construction Management (ACM):

In this approach, the **construction manager** acts as the **agent** for the **owner**, and all **contracts** are with the **owner**. This includes one or more contracts for **design services** and one or more contracts for **construction**. This arrangement utilizes the **talents, skills, and management abilities** of the selected **CM firm** in a strict, professional arrangement, which is useful for contracts in both the **private and public sectors**.

代理施工管理 (Agency Construction Management, ACM):

在这种方式中，**施工经理**作为**业主**的**代理人**，所有**合同**都由**业主**签订。这包括一个或多个**设计服务合同**和一个或多个**施工合同**。这种安排充分利用了所选**施工管理公司**的**才干、技能和管理能力**，是一种严谨的专业安排，适用于**私营和公共部门**的合同。

简化版:

Agency Construction Management (ACM)	代理施工管理 (ACM)
• Construction manager acts as owner's agent • Contracts are signed by the owner • Suitable for private and public sectors	• 施工经理为业主代理 • 合同由业主签订 • 私营和公共部门适用

(2) Guaranteed-Maximum-Price Construction Management (GMPCM):

This is a combination of **construction management** and **general contracting**, where the **owner** utilizes the **talents and skills** of the **construction manager** during the **conceptual, schematic, and design development** stages, and its **contracting ability** during **bidding and construction**. In this form of CM, the **owner** establishes **incentives** (penalties/rewards) with the **construction manager**, who then **bids** and issues **contracts to subcontractors**.

保证最高价施工管理 (Guaranteed-Maximum-Price Construction Management, GMPCM):

这是一种**施工管理**与**总承包**相结合的形式，其中**业主**在**概念设计、方案设计和设计开发阶段**利用**施工经理的才干和技能**，并在**投标和施工阶段**利用其**合同能力**。在这种**施工管理形式**下，**业主**与**施工经理**建立了**激励措施**（奖惩机制），**施工经理**负责**投标**并向**分包商**签订合同。

简化版:

Guaranteed-Maximum-Price Construction Management (GMPCM)	保证最高价施工管理 (GMPCM)
• Combines construction management and general contracting • Uses construction manager's skills in design phase • Incentive system (penalties/rewards)	• 施工管理和总承包结合 • 设计阶段利用施工经理的技能 • 激励机制（奖惩）

(3) Extended Service Construction Management:

This is a combination of **Agency** and **GMP** forms of CM, in which the **owner** allows the **construction manager** to participate in the **construction phase** through **bidding** or **negotiation**, in whole or in part.

延伸服务施工管理 (Extended Service Construction Management):

这是一种结合了代理施工管理 (ACM)和保证最高价 (GMP)形式的施工管理方式，业主允许施工经理通过**投标或谈判**（全部或部分）参与**施工阶段**。

简化版:

Extended Service Construction Management - Combines Agency and GMP forms Construction manager involved in construction phase	延伸服务施工管理 - 结合代理管理和保证最高价 - 施工经理参与施工阶段
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(4) Owner Construction Management:

In this form, the **owner** and its **in-house personnel** accept the **responsibility**, **benefits**, and **risks** of performing CM services. The **owner** employs a **staff** of **design** and **construction personnel** capable of handling all aspects of a project from **conceptual design** through **construction**. **Continuous construction activity** within the owner's organization is typically required for this form of CM to be **cost-effective**.

业主施工管理 (Owner Construction Management):

在这种形式下，业主及其内部人员承担了提供施工管理服务**的责任、利益和风险**。业主雇用具备处理从**概念设计**到**施工**各方面能力的设计和**施工人员**。通常，业主组织中需要**持续的施工活动**，以确保这种施工管理形式的**成本效益**。

简化版:

Owner Construction Management - Managed in-house by owner - Involves design through construction - Requires continuous construction for cost-effectiveness	业主施工管理 - 业主内部负责管理 - 设计到施工全程参与 - 需持续施工活动以节省成本
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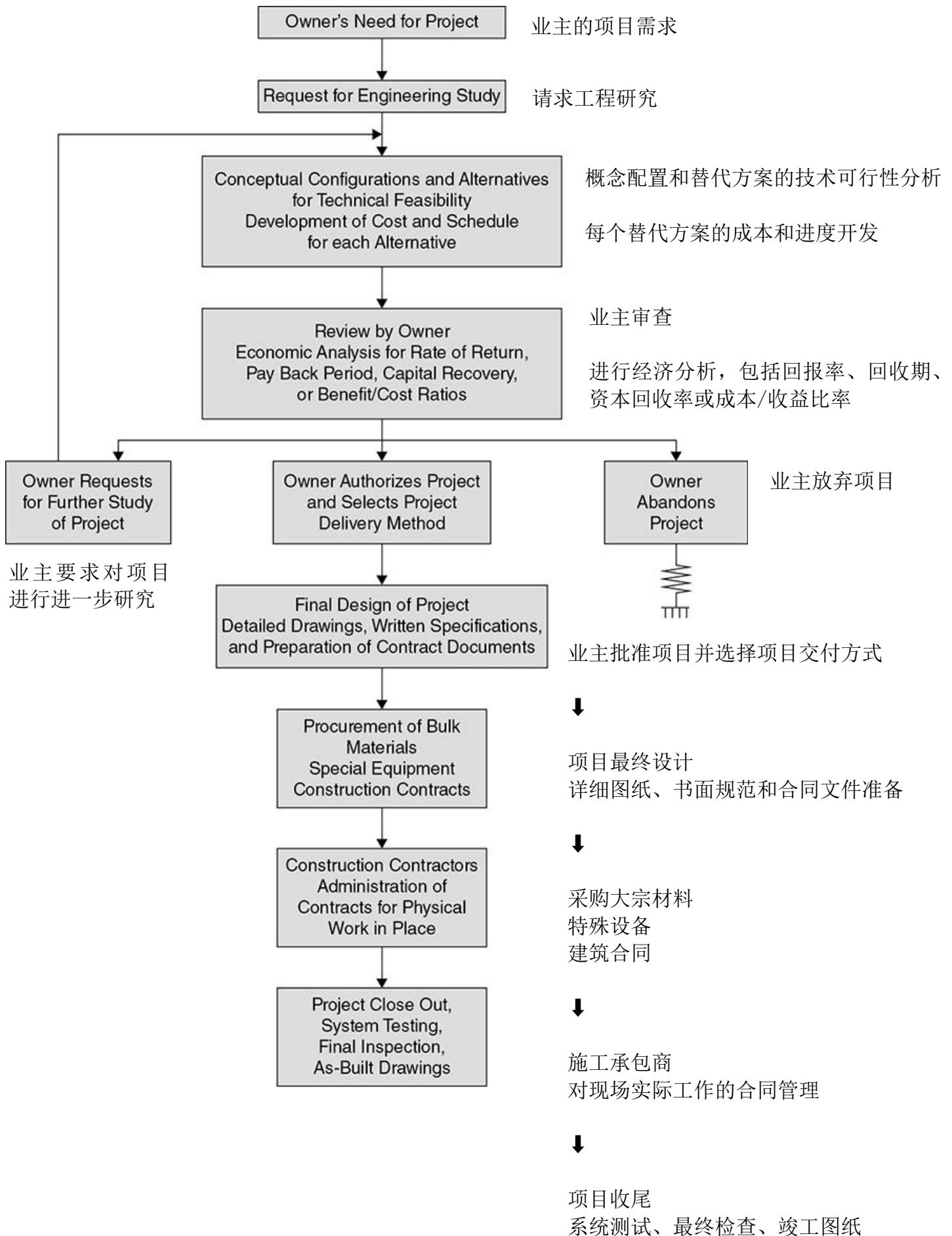
3. The various **phases** for the development of a project are shown in **Figure 3-2**. Review each phase and identify the **party** that would most likely be involved in performing the **work** of the phase, and the party whose work will be most influenced by the **results** of the phase. Identify parties as one of the following: **owner**, **consultant** to the owner, **designer**, **construction manager**, or **construction contractor**.

图 3-2 展示了项目开发的各个阶段。回顾每个阶段，并确定最可能参与该阶段工作的相关方，以及其工作的成果将对哪一方产生最大影响。相关方可分为以下几类：业主、业主的顾问、设计师、施工经理或建筑承包商。

Process of Project Development	Involvement	Affected Party
Owner's Need for Project	Owner's Staff	All Parties
Request for Engineering Study	Owner/Consultant	Designer
Conceptual Configurations/Alternatives	Designer/Owner/CM	Owner/Contractors
Review by Owner, Economic Analysis	Owner/CM	Owner
Final Design of Project, Plans/Specs	Designer/CM	Contractors/Owners
Procurement of Bulk Material/Equipment	Owner/Designer/CM	Contractors
Construction	Contractors	Owner/Designer/CM
Project Close Out	All Parties	All Parties

项目开发过程	参与方	受影响方
业主的项目需求	业主人员	所有方
工程研究请求	业主/顾问	设计方
概念配置/替代方案	设计方/业主/施工经理	业主/承包商
业主审核，经济分析	业主/施工经理	业主
项目最终设计、图纸/规格	设计方/施工经理	承包商/业主
采购大宗材料/设备	业主/设计方/施工经理	承包商
施工	承包商	业主/设计方/施工经理
项目收尾	所有方	所有方

FIGURE 3-2 Phases of a Project.



4. A part of the **owner's study** is defining the **requirements** of the project. Identify and describe **methods** that can be used to assist an owner with this important study, assuming that the owner does not have the **expertise** to perform the study.

业主的研究的一部分是确定项目的**需求**。识别并描述一些**方法**，帮助业主完成这项重要的研究，假设业主没有执行该研究的**专业知识**。

Solution:

- Meet with the **owner's staff**, particularly the **operations** and **maintenance personnel** who will be using the project upon completion.
 - Spend adequate time with the **operations/maintenance personnel** to understand the **work they do**, **how they do it**, and **their objectives**.
 - Conduct a series of meetings to distinguish between "**what they need**" and "**what they want**," and relate those needs to the **project requirements** that must be designed and built.
 - Develop the **preliminary scope, budget, and schedule requirements** of the project, and translate these requirements into terms that can be understood by the **owner's staff**.
 - Obtain **feedback** from the **owner's staff** to confirm their understanding of the project requirements necessary to meet the needs of the **owner's user group**.
 - Document the **results of meetings, discussions, alternatives, and decisions made**, and ensure that these results are distributed to the **design team**.
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- 与业主的工作人员会面，特别是将来会使用该项目的运营和维护人员。
 - 花足够的时间与运营/维护人员沟通，了解他们的工作内容、工作方式以及工作目标。
 - 进行一系列会议，将“他们需要的”与“他们想要的”区分开来，并将这些需求与项目中需要**设计和建造**的要求联系起来。
 - 制定项目的**初步范围、预算和进度要求**，并将这些要求转换为业主工作人员能够理解的术语。
 - 获取业主工作人员的反馈，确认他们理解为满足业主**用户群体需求**所需完成的项目要求。
 - 记录会议结果、讨论内容、替代方案和已做出的决定，并确保这些结果分发给**设计团队**。

简化版:

Methods to Assist the Owner in Defining Project Requirements:

- **Meet with owner's staff:** especially **operations and maintenance personnel**
- **Understand work and goals:** spend time communicating
- **Separate needs from wants:** clarify through meetings
- **Develop preliminary scope, budget, and schedule:** translate into owner's terms
- **Get feedback:** confirm owner's staff understands project needs
- **Document meeting results:** distribute to the **design team**

帮助业主确定项目需求的方法

- 与业主人员会面：特别是运营和维护人员
- 了解工作内容和目标：花时间沟通
- 区分需求与想要：通过会议厘清
- 制定初步范围、预算和进度：转换为业主理解的术语
- 获取反馈：确认业主人员理解项目需求
- 记录会议结果：分发给设计团队

5. Review the list of **references** at the end of this chapter and summarize the process of **setting project objectives** to assure the **successful completion** of a project.

查看本章末尾的**参考文献**列表，总结**设定项目目标**的过程，以确保项目能够**成功完成**。

Solution:

- A **needs assessment** that identifies the **owner's justification** to proceed with the project.
 - The **goals and objectives** of the group within the **owner's organization** that will use the project.
 - A **list of needs** that must be met to satisfy the **end users** of the project.
 - **Budget and schedule constraints** established by the owner.
 - The **minimum requirements** of **quality and performance**.
 - The **maximum permissible budget** that can be expended for the project.
 - The required **completion date**, or "**need date**," for the project.
 - **Safety requirements** of the owner and **governmental agencies**.
 - **Environmental requirements** of the owner and **government agencies**.
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- 需求评估，确定业主推进项目的理由。
 - 将要使用该项目的业主组织内部门的目标和宗旨。
 - 满足项目**最终用户**的需求清单。
 - 业主设定的**预算和进度**限制。
 - **质量和性能**的最低要求。
 - 项目的**最高可支出预算**。
 - 项目所需的**完成日期**或“**需求日期**”。
 - 业主和政府机构的**安全要求**。
 - 业主和政府机构的**环境要求**。

简化版:

Process of Setting Project Objectives

- **Needs assessment:** reason for owner to start the project
- **Goals of user groups:** objectives of departments using the project
- **User needs list:** requirements to satisfy end users
- **Budget and schedule:** constraints set by owner
- **Minimum quality requirements:** basic quality and performance standards
- **Maximum budget:** project's spending limit
- **Completion date:** set the "need date"
- **Safety requirements:** standards from owner and government
- **Environmental requirements:** comply with owner and government regulations

设定项目目标的过程

- | | |
|---|---|
| <ul style="list-style-type: none">• 需求评估：确定业主启动项目的理由• 使用部门的目标：明确用户目标和宗旨• 用户需求清单：满足最终用户需求• 预算和进度：业主设定的限制• 最低质量要求：保证基本质量和性能 | <ul style="list-style-type: none">• 最高预算：项目的最大支出限额• 完成日期：确定“需求日期”• 安全要求：业主和政府的标准• 环境要求：符合业主和政府规定 |
|---|---|

6. Describe the **differences** between **project objectives** and **project scope definition**. Define the **interactive role** of the **owner** and **designer** in this process to ensure a **well-defined understanding** between the two parties.

描述项目目标与项目范围定义之间的差异。定义业主和设计师在此过程中的互动角色，以确保双方达成明确的理解。

Solution:

(1) Project Objectives:

- The **owner** has the primary responsibility for setting **project objectives**.
- Project objectives define the **needs of the owner** that must be met by the project and can include **economic** or **financial results**.
- A detailed definition of project objectives is a **prerequisite** to defining the **project scope**.
- There needs to be a **separation between "what is needed" and "what is wanted"**.

项目目标:

业主对设定项目目标负有主要责任。

项目目标定义了**业主的需求**，项目必须满足这些需求，并且可能包括**经济或财务结果**。

对项目目标的详细定义是确定**项目范围的前提条件**。

需要区分“需要的”和“想要的”内容。

简化版:

Project Objectives:

- **Owner's responsibility:** sets objectives
- **Defines needs:** meets owner's economic or financial goals
- **Prerequisite:** clear objectives before defining scope
- **Separate** “needs” from “wants”

项目目标 (Project Objectives):

- **业主责任:** 设定目标
- **定义需求:** 满足业主经济或财务结果
- **前提条件:** 目标清晰后才能定义范围
- **区分“需要”和“想要”**

(2) Project Scope:

- The **designer** has the primary responsibility for defining the **scope** of the project.
- The project scope identifies **work items** or "**deliverables**" required to achieve the project.
- Each **work item** in the scope should be **budgeted** and **scheduled** before starting work.
- There needs to be a **separation between "what can be" and "what can't be" accomplished**.

项目范围:

- 设计师对定义项目的**范围**负有主要责任。
- 项目范围识别了为实现项目所需的工作项或“交付成果”。
- 在开始工作之前，每个工作项都应进行**预算**和**进度安排**。
- 需要区分“可以完成的”和“无法完成的”内容。

简化版:

Project Scope:	项目范围 (Project Scope):
• Designer's responsibility: defines scope • Work items/deliverables: identifies what's needed for the project • Budget and schedule: plan work items beforehand • Separate "what can be done" from "what can't"	• 设计师责任: 定义范围 • 工作项/交付成果: 识别项目所需 • 预算和进度: 工作项须先安排 • 区分“能完成”和“无法完成”

(3) Interactive Roles:

- A **joint meeting** of the **owner** and **designer** is necessary to clarify **objectives** and **scope**.
- The **owner** should explain objectives in terms that can be understood by the **designer**.
- The **designer** should explain scope in terms that can be understood by the **owner**.
- Both the **owner** and **designer** should agree that the **objectives** and **scope** are compatible.

互动角色:

- 需要业主和设计师的**联合会议**，以明确目标和范围。
- 业主应以设计师能够理解的术语阐述项目目标。
- 设计师应以业主能够理解的术语阐述项目范围。
- 业主和设计师都应确保项目目标和项目范围一致、相互兼容。

简化版:

Interactive Roles:	互动角色 (Interactive Roles):
• Joint meeting: clarify objectives and scope • Owner: explains objectives in designer's terms • Designer: explains scope in owner's terms • Both confirm: ensure objectives and scope align	• 联合会议: 明确目标和范围 • 业主: 用设计师理解的术语解释目标 • 设计师: 用业主理解的术语解释范围 • 双方确认: 确保目标与范围一致

7. Review **published articles** that describe the concept and process of **partnering**, and list **major differences** between partnering and **traditional methods** of **contracting strategies**.

查阅描述合作概念和过程的已发表文章，列出合作与传统合同策略之间的主要区别。

Solution:

(1) Traditional Contracting Strategies:

- Generally, there is an attempt to allocate **risks, responsibilities**, and **authority** among parties.
- Projects are often either **cost** or **schedule-driven**, or sometimes both.
- The **contracts** between parties are **short-term**, only for the duration of each project.

传统合同策略:

- 通常尝试在各方之间分配**风险、责任和权力**。
- 项目通常以**成本或进度**为驱动，有时兼顾两者。
- 各方之间的**合同为短期**，仅限于每个项目的持续时间。

简化版:

Traditional Contracting Strategies: • Allocate: risks, responsibilities, authority • Driven by: cost or schedule • Short-term contracts: for each project	传统合同策略 (Traditional Contracting Strategies): • 分配: 风险、责任、权力 • 驱动因素: 成本或进度 • 短期合同: 每个项目单独签订
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(2) Partnering Strategies:

- Generally, there is an attempt to **share risks, responsibilities**, and **authority** between parties.
- Information on **costs** and **schedules** of projects is jointly **shared** and **agreed upon**.
- The **arrangements** between parties are **long-term** rather than short-term.

合作伙伴策略:

- 通常尝试在各方之间**共享 风险、责任和权力**。
- 项目的**成本和进度**信息由各方共同分享并达成一致。
- 各方之间的**安排是长期的**，而非短期的。

简化版:

Partnering Strategies: • Share: risks, responsibilities, authority • Transparency: jointly share cost and schedule information • Long-term cooperation: rather than short-term	合作伙伴策略 (Partnering Strategies): • 共享: 风险、责任、权力 • 信息透明: 共同分享成本和进度信息 • 长期合作: 而非短期
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